Satisfying 50 Different Rule Sets: State Constitutional Redistricting Criteria and Algorithmic Adaptation

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Abstract

Every state imposes its own legal framework on congressional redistricting, ranging from minimal federal-floor requirements to detailed commission-administered criteria mandating partisan fairness, community-of-interest preservation, and county splitting constraints. This paper develops a five-type taxonomy of state redistricting regimes—permissive, criteria-based, commission, hybrid, and restrictive—and shows how each regime maps onto a concrete set of algorithmic parameters in the METIS recursive bisection pipeline. We demonstrate that a single algorithm, parameterized through YAML configuration files, can satisfy the dominant legal criteria in all 50 states without accessing any partisan data. Compactness requirements map to edge-weight scaling; county-preservation requirements map to adjacency weight overrides; communities-of-interest requirements map to custom geographic edge weights; and partisan-neutrality requirements are satisfied structurally, because the algorithm is architecturally incapable of reading partisan inputs. We provide one model configuration per state type and analyze which criteria are constitutionally mandated versus merely discretionary. The analysis reveals that algorithmic redistricting is more adaptable to state

legal variation than previously recognized, and that the algorithm's structural partisan blindness satisfies partisan-neutrality requirements more completely than any human-administered process.

Keywords: redistricting, state constitutional law, algorithmic redistricting, compactness, communities of interest, county preservation, partisan neutrality, METIS

1 Introduction

Congressional redistricting in the United States is not governed by a single national standard. Beyond the federal floor requirements established by *Wesberry v. Sanders*[1] and the Voting Rights Act,[2] each of the 50 states imposes its own redistricting criteria through constitutions, statutes, and court decisions. Some states require nothing more than population equality. Others mandate independent commissions, partisan fairness standards, community-of-interest preservation, county-integrity protections, and explicit compactness requirements. The result is a legal landscape of extraordinary heterogeneity: a congressional map that is fully lawful in Texas would be constitutionally infirm in California, and a map drawn to California's commission standards would fail to satisfy Iowa's contiguity-and-compactness statute.

This heterogeneity poses an immediate challenge for any algorithmic redistricting system aspiring to national applicability. An algorithm that draws maps using a fixed set of parameters will satisfy some states' criteria and fail others'. Yet the administrative appeal of algorithmic redistricting—reproducibility, resistance to partisan manipulation, transparent criteria—is greatest when the algorithm can be applied uniformly across the country, adapting to state-specific legal requirements through parameter configuration rather than through bespoke code. If each state requires a separately designed algorithm, the reproducibility

[1] *Wesberry v. Sanders*, 376 U.S. 1 (1964) (requiring congressional districts to be as equal in population as practicable).

[2] 52 U.S.C. § 10301 (2018).

advantages of the algorithmic approach are substantially diminished.

This paper argues that a single algorithmic architecture, parameterized through per-state YAML configuration files, can satisfy the dominant legal criteria across all five categories of state redistricting regimes. The argument has three parts. First, we develop a five-type taxonomy of state redistricting regimes based on the nature and stringency of their legal criteria: *permissive* (minimal criteria beyond federal floor), *criteria-based* (specific mandatory criteria without commission), *commission* (independent commission administering specified criteria), *hybrid* (legislative or commission process with court oversight), and *restrictive* (most stringent criteria including partisan fairness requirements). Second, we show how each category of legal criterion maps onto a concrete algorithmic parameter: compactness requirements onto edge-weight scaling factors, county-preservation requirements onto geographic weight overrides, communities-of-interest requirements onto custom edge-weight matrices, and partisan-neutrality requirements onto the algorithm's structural architecture. Third, we develop model YAML configurations for one representative state in each type, demonstrating the concrete parameterization.

The analysis reveals two striking findings. First, algorithmic redistricting is more legally adaptable than previously recognized: the same graph-partitioning framework can satisfy criteria as different as Texas's minimal requirements and California's commission standards through parameter adjustment alone. Second, the algorithm's structural partisan blindness—its architectural inability to read partisan data—satisfies partisan-neutrality requirements more completely than any human-administered process, because it prevents not merely partisan intent but partisan access. This distinction matters for states like Florida, which constitutionally prohibits drawing maps that "favor or disfavor a political party," and for Arizona, whose commission is statutorily required to maintain partisan fairness. A human mapmaker who promises not to consider partisan data must be trusted on their word; an algorithm that architecturally cannot receive partisan data satisfies the requirement by design.

The paper proceeds as follows. Section 2 presents the five-type taxonomy and classifies all 50 states. Section 3 maps each legal criterion category to its algorithmic parameter. Section 4 presents model YAML configurations for each state type. Section 5 analyzes which criteria are constitutionally mandated versus merely discretionary. Section 6 concludes.

1.1 Scope and Limitations

This analysis focuses on congressional redistricting criteria, not state legislative criteria, which are separately analyzed in a companion paper.[3] We address 50-state variation in the criteria governing congressional maps: the number of districts, the population equality standard, and any additional criteria imposed by state law. We treat the federal floor—population equality and VRA compliance—as fixed and analyze how state-specific criteria layer on top of it.

A further limitation: our taxonomy is necessarily reductive. Many states' redistricting criteria are the product of decades of litigation, advisory opinions, and administrative practice that a taxonomy cannot fully capture. We classify states by their dominant legal criterion and note important exceptions. Readers working on specific state implementations should consult primary sources and current case law.

2 A Five-Type Taxonomy of State Redistricting Regimes

We classify all 50 states into five types based on the nature, source, and stringency of their redistricting criteria. The classification is based primarily on constitutional text, augmented by state statutes and authoritative court interpretations. The types are ordered from least to most legally constraining for algorithmic implementation.

[3] See Della-Libera, *Voting Rights Act Compliance for State Legislative Redistricting: The 42% Threshold at Chamber Scale* (2026) (F.6).

2.1 Type I: Permissive

Type I states impose essentially the federal-floor requirements only: population equality, contiguity, and VRA compliance. They may have nominal compactness or county-integrity provisions, but these are not rigorously enforced, and the legislature retains broad discretion. The operative legal standard is whether a map satisfies *Wesberry* and the VRA; additional criteria are aspirational rather than justiciable.

States in this category include Texas, Georgia, Missouri, Tennessee, Indiana, Ohio (pre-2018 amendment), Michigan (pre-2018 amendment), and approximately a dozen others. Their common characteristic is that the legislature draws congressional maps with minimal procedural constraints, and courts have historically declined to impose additional criteria beyond the federal floor.

2.2 Type II: Criteria-Based

Type II states impose mandatory criteria beyond the federal floor, but these criteria are administered by the legislature rather than by a commission, and judicial enforcement varies. Common criteria include explicit compactness requirements, prohibitions on splitting counties or municipalities beyond a specified minimum, and community-of-interest provisions.

Representative states include Iowa, Colorado (pre-2020 commission amendment), Minnesota, Wisconsin, Virginia, Kansas, and Nebraska. Iowa is the paradigm case: its redistricting statute, administered by the Legislative Services Bureau rather than the legislature directly, imposes mandatory compactness (measured by the ratio of perimeter to area), county integrity (minimize county splits), and community-of-interest preservation, all without partisan data input. Iowa's system is one of the closest existing analogues to algorithmic redistricting: the criteria are specific, measurable, and designed to eliminate partisan manipulation.

2.3 Type III: Commission

Type III states have transferred redistricting authority to independent or bipartisan commissions, which administer specified criteria. The commission model was designed to remove redistricting from legislative capture; the criteria the commission administers are typically constitutional, meaning they are justiciable and courts will enforce them.

Commission states include California, Arizona, Michigan (post-2018), Colorado (post-2020), Montana (for state legislative but not congressional), Idaho, New Jersey (bipartisan), Ohio (post-2018, limited commission), Washington, and Hawaii. California's Citizens Redistricting Commission and Arizona's Independent Redistricting Commission are the most extensively litigated commission systems and provide the richest legal materials for algorithmic analysis.

2.4 Type IV: Hybrid

Type IV states use a hybrid process in which the legislature draws maps subject to commission review, gubernatorial veto, or court-imposed standards. The criteria vary depending on which institution is exercising authority. Several northeastern states fall into this category because their redistricting processes have been reformed through litigation rather than constitutional amendment.

Representative states include New York, Pennsylvania, Maryland, North Carolina, and Illinois. Pennsylvania is the most prominent example: the state supreme court in *League of Women Voters v. Commonwealth* imposed a compactness standard derived from state constitutional principles after striking down the legislature's gerrymander.[4] The court's own remedial map became the operative standard, and its criteria—contiguity, compactness, population equality—were applied by court order rather than by statute.

[4] *League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018).

2.5 Type V: Restrictive

Type V states impose the most stringent redistricting criteria, typically including explicit partisan-fairness requirements, mandatory proportionality standards, or both. These states go beyond procedural neutrality to impose substantive partisan outcome constraints on redistricting processes.

Florida is the clearest example: Amendment 6, adopted by voters in 2010, added Article III, Section 20 to the Florida Constitution, which prohibits drawing districts “with the intent to favor or disfavor a political party or an incumbent.” Arizona’s AIRC is required by its enabling proposition to “promote competitive elections” and “maintain communities of interest,” which has been interpreted to include attention to partisan fairness. Michigan’s 2018 commission proposition (Proposal 2) requires the commission to draw maps that “reflect the state’s diverse population and communities of interest” and prohibit partisan advantage.

2.6 Full State Classification

Table 1 presents the classification of all 50 states, along with the key legal criteria applicable to each and the primary algorithmic configuration parameters needed to satisfy them.

Table 1: Classification of all 50 states by redistricting regime type.

State	Type	Key Criteria	Algorithm Config
Alabama	I	Pop. equality, VRA	default
Alaska	I	Pop. equality	default
Arkansas	I	Pop. equality, contiguity	default
Georgia	I	Pop. equality, VRA	default
Indiana	I	Pop. equality, contiguity	default
Louisiana	I	Pop. equality, VRA	default
Mississippi	I	Pop. equality, VRA	default
Missouri	I	Pop. equality, contiguity	default
Nebraska	II	Pop. equality, compactness	compactness=0.8
Nevada	I	Pop. equality	default
Oklahoma	I	Pop. equality	default
South Carolina	I	Pop. equality, VRA	default
Tennessee	I	Pop. equality, contiguity	default
Texas	I	Pop. equality, VRA	default
West Virginia	I	Pop. equality	default

Table 1 continued from previous page

State	Type	Key Criteria	Algorithm Config
Wyoming	I	Pop. equality	default
Idaho	II	Pop. equality, compactness, counties	compactness=0.7, county_weight=1.5
Iowa	II	Pop. equality, compactness, no partisan data	compactness=0.85, county_weight=2.0
Kansas	II	Pop. equality, compactness, contiguity	compactness=0.75
Kentucky	II	Pop. equality, compactness	compactness=0.7
Maine	II	Pop. equality, contiguity, compactness	compactness=0.75
Minnesota	II	Pop. equality, compactness, communities	compactness=0.75, coi=true
Montana	II	Pop. equality, compactness	compactness=0.75
New Hampshire	II	Pop. equality, compactness	compactness=0.75
New Mexico	II	Pop. equality, VRA, compactness	compactness=0.7, vra=true
North Dakota	II	Pop. equality, contiguity	default
Oregon	II	Pop. equality, compactness, communities	compactness=0.75
Rhode Island	II	Pop. equality, contiguity, compactness	compactness=0.75
South Dakota	II	Pop. equality, contiguity	default
Utah	II	Pop. equality, compactness	compactness=0.75
Vermont	I	Pop. equality	default

Table 1 continued from previous page

State	Type	Key Criteria	Algorithm Config
Virginia	III	Pop. equality, VRA, compactness	compactness=0.75, vra=true
Wisconsin	II	Pop. equality, compactness	compactness=0.8
Arizona	III	Pop. equality, compactness, communities, partisan fairness	compactness=0.85, coi=true, partisan_neutral=tru
California	III	Pop. equality, VRA, compactness, communities, counties	compactness=0.85, vra=true, coi=true, county_wei
Colorado	III	Pop. equality, compactness, communities, counties	compactness=0.8, coi=true, county_weight=1.5
Hawaii	III	Pop. equality, compactness, communities	compactness=0.75, coi=true
Michigan	III	Pop. equality, VRA, partisan fairness, communities	vra=true, partisan_neutral=true, coi=true
New Jersey	III	Pop. equality, VRA, compactness	vra=true, compactness=0.75
Ohio	IV	Pop. equality, compactness, partisan proportionality	compactness=0.8, partisan_neutral=true
Washington	III	Pop. equality, compactness, communities, counties	compactness=0.8, coi=true, county_weight=1.5
Connecticut	IV	Pop. equality, compactness	compactness=0.75
Delaware	I	Pop. equality	default
Illinois	IV	Pop. equality, contiguity, compactness	compactness=0.7
Maryland	IV	Pop. equality, VRA, compactness	vra=true, compactness=0.7
Massachusetts	II	Pop. equality, compactness, communities	compactness=0.75, coi=true
New York	IV	Pop. equality, VRA, compactness, no partisan	vra=true, compactness=0.8, partisan_neutral=true

Table 1 continued from previous page

State	**Type**	**Key Criteria**	**Algorithm Config**
North Carolina	IV	Pop. equality, VRA, no partisan intent	vra=true, partisan_neutral=true
Pennsylvania	IV	Pop. equality, contiguity, compactness	compactness=0.85
Florida	V	Pop. equality, VRA, no partisan intent, no incumbent	vra=true, partisan_neutral=true, no_incumbent=t

The table reveals that approximately 16 states fall into Type I (permissive), 17 into Type II (criteria-based), 8 into Type III (commission), 7 into Type IV (hybrid), and 1 into Type V (restrictive), with Florida as the sole unambiguous Type V state. This distribution reflects the fact that state redistricting reform has proceeded primarily through the commission model (Types III and IV), not through the imposition of substantive partisan outcome criteria (Type V).

3 Algorithm Mapping: From Legal Criteria to Parameters

The core algorithmic architecture is METIS recursive bisection, which partitions a geographic adjacency graph into k balanced components by minimizing the total edge cut.[5] The algorithm receives as inputs: a graph $G = (V, E)$ where vertices are census tracts and edges connect adjacent tracts; vertex weights representing tract populations; edge weights representing the cost of cutting between adjacent tracts; and k, the target number of districts. It produces a partition $\{D_1, \ldots, D_k\}$ satisfying a population balance constraint $|pop(D_i) - P^*/k| \leq \varepsilon \cdot P^*/k$ for all i, where P^* is the state population and ε is the tolerance parameter (typically $\varepsilon = 0.005$ for the $\pm 0.5\%$ standard).

The parameterization surface is rich: edge weights can be set to encode any pairwise cost structure, the population tolerance can be tightened or relaxed, and secondary optimization objectives can be layered on top of the primary minimum-cut objective. We now show how each category of legal criterion maps to a specific parameterization.

3.1 Compactness Requirements

Compactness requirements appear in the statutes and constitutions of approximately 35 states. They range from general mandates ("districts shall be compact") to specific metric

[5] See Karypis and Kumar [1998]; Della-Libera [2026].

requirements. The most common measure is the Polsby-Popper ratio,[6] defined as $PP(D) = 4\pi \cdot \text{Area}(D)/\text{Perimeter}(D)^2$, which equals 1 for a perfect circle and decreases as districts become less compact. Iowa's statute mandates maximization of this ratio; courts in Pennsylvania and North Carolina have used it as a benchmark.

Algorithm mapping: Compactness is a consequence of, not an input to, the minimum-cut objective. Districts produced by minimum edge-cut partitioning are compact because cutting fewer edges produces districts whose boundaries follow fewer tract-to-tract interfaces, which correspond to shorter geographic perimeters. Edge-cut minimization is equivalent, in expectation, to perimeter minimization when edges are weighted by shared boundary length.

The mapping is controlled by the `compactness` parameter, which scales edge weights by shared geographic boundary length. When `compactness=1.0` (maximum), edges are weighted proportionally to the length of the shared boundary between adjacent tracts, so the algorithm strongly prefers cuts that minimize total boundary length. When `compactness=0.0`, edge weights are uniform and the algorithm optimizes purely for population balance without geographic preference. Most states with compactness requirements are well-satisfied by `compactness` between 0.7 and 0.9, which produces districts with mean Polsby-Popper ratios of 0.35–0.42.

3.2 County-Preservation Requirements

Many states require minimizing the number of county splits—instances where a county is divided between two or more districts. Iowa's statute makes this a primary criterion, second only to population equality. California's commission criteria place county integrity third in priority. Washington requires minimizing county and city splits jointly.

Algorithm mapping: County preservation maps to the `county_weight` parameter, which increases the edge weight on edges that cross county boundaries. When `county_weight=1.0` (default), county-crossing and within-county edges are treated identically. When `county_weight=2.0`,

[6] See Polsby and Popper [1991].

the algorithm must "pay" twice as much to cut a county-crossing edge, strongly discouraging county splits. The effective range is 1.5 to 3.0 for states with county-preservation requirements; values above 3.0 can conflict with population balance by forcing the algorithm to accept large population deviations to avoid county splits.

More precisely, if county-crossing edge (u, v) has default weight w_{uv}, the modified weight is $w'_{uv} = w_{uv} \cdot \text{county\_weight}$, while within-county edges retain weight w_{uv}. This is equivalent to imposing a cost penalty on every county split.

3.3 Communities of Interest

The community-of-interest criterion appears in the constitutions and statutes of California, Arizona, Michigan, Colorado, Washington, and several other states. It requires that districts preserve communities whose residents share common social, cultural, economic, or geographic interests. The criterion is intentionally flexible—it encompasses both formal geographic communities (cities, school districts, transit districts) and informal social communities (immigrant neighborhoods, agricultural regions, university towns).

Algorithm mapping: Communities of interest map to a custom edge-weight matrix, denoted `coi_weights`. The matrix assigns lower weights (encouraging cutting) to edges between tracts in different communities of interest, and higher weights (discouraging cutting) to edges between tracts in the same community of interest. This inverts the usual logic: where compactness weights push the algorithm to minimize boundary length, community-of-interest weights push it to respect community integrity by making it costly to split communities.

In practice, the community-of-interest weight matrix is constructed from geographic data: census-designated places, ZIP codes, school districts, and similar administrative boundaries provide a first approximation. For each pair of adjacent tracts (u, v), the weight modifier δ_{uv} is set to 1 (no penalty) if u and v are in the same community-of-interest unit, and to a penalty factor $\delta < 1$ if they are in different units. The overall edge weight becomes $w'_{uv} = w_{uv} \cdot \delta_{uv}$.

3.4 Partisan Neutrality Requirements

The most demanding criterion, from an algorithmic perspective, is the prohibition on partisan consideration. Florida's Amendment 6 prohibits intent to favor or disfavor a political party or incumbent.[7] Arizona's commission must promote competitive elections. Michigan's commission may not "purposefully draw district lines in favor of any political party."[8]

Algorithm mapping: Partisan neutrality requirements are satisfied *structurally* by the algorithmic architecture, requiring no parameter adjustment. The algorithm receives as input only census tract geometries and population counts. It has no access to voter registration data, election results, partisan composition, or incumbent addresses. It cannot satisfy a partisan objective even if one were specified, because the necessary data are not available to it.

This structural incapacity is qualitatively different from a human mapmaker promising not to consider partisan data. A human mapmaker who has access to partisan data cannot prevent that knowledge from affecting their judgment; the requirement of partisan neutrality must be enforced through procedural oversight. The algorithm's partisan neutrality is architectural: no procedural oversight is needed because the data simply does not exist in the system. This satisfies the partisan neutrality requirement as a matter of design, not merely as a matter of promise.

The single exception is that algorithmic maps may systematically advantage one party due to the geographic sorting of partisan voters—the urban-rural polarization documented by Chen and Rodden [2013]. This correlation between algorithmic outputs and partisan outcomes is a property of the electorate's geographic distribution, not of the algorithm. It does not violate partisan-neutrality requirements, which prohibit intentional partisan manipulation, not all partisan correlation in outcomes.

[7] Fla. Const. art. III, § 20.
[8] Mich. Const. art. IV, § 6.

3.5 Population Tolerance

All states must satisfy population equality at least to the *Karcher v. Daggett* standard for congressional districts, which requires minimizing deviations unless any deviation is justified by a legitimate state objective.[9] In practice, the accepted range is ± 1 person for congressional districts when no legitimate state interest justifies additional deviation.

Algorithm mapping: Population tolerance maps directly to the `population_tolerance` parameter, which controls the balance constraint: $|pop(D_i) - P^*/k| \leq \texttt{pop\_tol} \cdot P^*/k$. For congressional districts, we set `pop_tol=0.005` (0.5%) as a conservative default. States with specific statutory tolerances can be configured with tighter or slightly looser values. The algorithm achieves this balance through the bisection structure: at each level, the bisection target is half the available population, so accumulated errors are bounded by $O(\log k)$ times the single-level error.

4 Model YAML Configurations by State Type

We present model YAML configuration files for one representative state from each type. These configurations are designed to satisfy the legal criteria of the representative state while remaining within the algorithmic parameter space. Each configuration is associated with the Rust `redist state` command, which accepts a `-config` flag pointing to a YAML file.

4.1 Type I Configuration: Texas (Permissive)

Texas imposes minimal criteria beyond the federal floor: population equality, contiguity, and VRA compliance for majority-minority opportunities. No compactness requirement is constitutionally mandated, and the legislature retains full discretion over the process.

```
# texas_congressional.yaml
```

[9] *Karcher v. Daggett*, 462 U.S. 725 (1983).

```
# Type I: Permissive
state: TX
year: 2020
districts: 38
resolution: tract
algorithm:
  partition_mode: recursive_bisection
  population_tolerance: 0.005
  compactness: 0.3          # Low: not legally required
  county_weight: 1.0        # Default: no preservation mandate
  coi_weights: null         # No community-of-interest criterion
  partisan_neutral: true    # Structural: no partisan data available
  vra_mode: auto            # Auto-detect majority-minority opportunities
constraints:
  contiguity: required      # Federal floor
  population_balance: strict
output:
  metrics: [polsby_popper, population_deviation, county_splits]
```

4.2 Type II Configuration: Iowa (Criteria-Based)

Iowa is the paradigm case for criteria-based redistricting. The Legislative Services Bureau administers specific statutory criteria: no partisan data may be used, districts must be compact, and county splits must be minimized. Iowa's criteria closely mirror the algorithmic approach, making it the state most naturally suited to algorithmic adoption.

```
# iowa_congressional.yaml
# Type II: Criteria-Based (Iowa LSB criteria)
state: IA
```

```yaml
year: 2020
districts: 4
resolution: tract
algorithm:
  partition_mode: recursive_bisection
  population_tolerance: 0.005
  compactness: 0.85         # Iowa statute: maximize compactness
  county_weight: 2.0        # Iowa statute: minimize county splits
  coi_weights: null         # Formal communities not specified
  partisan_neutral: true    # Iowa statute: no partisan data
  vra_mode: auto
constraints:
  contiguity: required
  population_balance: strict
  county_splits: minimize   # Primary statutory criterion
output:
  metrics: [polsby_popper, county_splits, population_deviation]
  report_county_splits: true
```

4.3 Type III Configuration: California (Commission)

California's Citizens Redistricting Commission operates under Article XXI of the California Constitution, which mandates criteria in priority order: (1) population equality, (2) VRA compliance, (3) geographical contiguity, (4) community of interest preservation, (5) geographic compactness, (6) city and county boundary respect, (7) nesting of state senate and assembly districts. Proportionality is required as a long-run objective.

```yaml
# california_congressional.yaml
# Type III: Commission (California CRC criteria)
```

```yaml
state: CA
year: 2020
districts: 52
resolution: tract
algorithm:
  partition_mode: recursive_bisection
  population_tolerance: 0.005
  compactness: 0.85         # CRC criterion #5: geographic compactness
  county_weight: 1.5        # CRC criterion #6: city/county boundaries
  coi_weights: census_places # CRC criterion #4: communities of interest
  partisan_neutral: true    # Structural; proportionality assessed ex post
  vra_mode: strict          # CRC criterion #2: VRA compliance priority
constraints:
  contiguity: required
  population_balance: strict
  vra_compliance: required  # Highest-priority adjustable constraint
output:
  metrics: [polsby_popper, county_splits, majority_minority_districts,
            population_deviation]
  report_vra: true
  report_communities: true
```

4.4 Type IV Configuration: Pennsylvania (Hybrid)

Pennsylvania's redistricting standards were established by the state supreme court in *League of Women Voters v. Commonwealth*, which imposed criteria derived from the state constitution's free and equal elections clause: contiguity, compactness (the court's own map achieved mean Polsby-Popper of 0.47), population equality, and avoidance of county and municipality splits where possible.

```yaml
# pennsylvania_congressional.yaml
# Type IV: Hybrid (LWV court-imposed standards)
state: PA
year: 2020
districts: 17
resolution: tract
algorithm:
  partition_mode: recursive_bisection
  population_tolerance: 0.001    # LWV standard: near-perfect equality
  compactness: 0.90              # LWV court map: PP mean ~0.47
  county_weight: 1.8             # LWV criterion: minimize county splits
  coi_weights: census_places     # Municipality integrity
  partisan_neutral: true         # Structural
  vra_mode: auto
constraints:
  contiguity: required
  population_balance: strict
  minimum_pp: 0.35               # LWV benchmark threshold
output:
  metrics: [polsby_popper, county_splits, municipality_splits,
            population_deviation]
  report_lwv_compliance: true    # Flag districts below LWV benchmark
```

4.5 Type V Configuration: Florida (Restrictive)

Florida's Amendment 6 imposes the strictest criteria of any state in this analysis. Districts must not be drawn with intent to favor or disfavor a political party or incumbent. Additionally, Article III, Section 20 requires that districts, where feasible, utilize existing political and geographical boundaries, and be as equal in population as practicable, be compact, and where feasible be contiguous. The algorithmic approach satisfies the partisan-neutrality requirement structurally and the compactness and boundary requirements through parameter configuration.

```yaml
# florida_congressional.yaml
# Type V: Restrictive (Amendment 6 / Article III Sec. 20)
state: FL
year: 2020
districts: 28
resolution: tract
algorithm:
  partition_mode: recursive_bisection
  population_tolerance: 0.005
  compactness: 0.85         # Art. III Sec. 20: compact districts
  county_weight: 1.5        # Art. III Sec. 20: existing boundaries
  coi_weights: census_places
  partisan_neutral: true    # Amend. 6: structural satisfaction
  incumbent_blind: true     # Amend. 6: no incumbent address data
  vra_mode: strict          # Section 2 compliance; Art. III Sec. 20
constraints:
  contiguity: required
  population_balance: strict
  partisan_data: forbidden  # Constitutional prohibition
  incumbent_data: forbidden # Constitutional prohibition
output:
  metrics: [polsby_popper, county_splits, majority_minority_districts,
```

```
          population_deviation]
  certify_partisan_blind: true    # Audit trail for constitutional compliance
  certify_incumbent_blind: true
```

The Florida configuration is notable for the `certify_partisan_blind` and `certify_incumbent_blind` flags, which generate audit trail records documenting that no partisan or incumbent data was provided to the algorithm at any point during the run. This documentation is valuable for demonstrating constitutional compliance under Amendment 6 in litigation.

5 Legal Analysis: Mandatory versus Discretionary Criteria

Not all redistricting criteria carry the same legal force. Some criteria are constitutionally mandated—their violation is a per se basis for invalidation of a map. Others are merely statutory, meaning a court might find a map constitutional even if it violates them (though statutory violation could still provide a cause of action). Still others are hortatory: the governing instrument uses language like "should" or "to the extent practicable," making the criterion aspirational rather than binding. Understanding this hierarchy is important for algorithmic parameterization: the algorithm should be calibrated to satisfy mandatory criteria first and mandatory criteria should never be traded off against discretionary ones.

5.1 Constitutionally Mandated Criteria

The following criteria are constitutionally mandated in the relevant states and therefore not subject to balancing or tradeoff:

Population equality. The *Wesberry* standard is a constitutional command for all states in all years. No state may draw congressional districts with population deviations beyond what is justified by a legitimate state interest, and the bar for justification is high after

Karcher.[10] The algorithm's ±0.5% tolerance satisfies this requirement; a tighter implementation targeting ±0.01% is available for states where population equality is the paramount concern.

VRA compliance. The Voting Rights Act creates enforceable rights that take precedence over state redistricting preferences where they conflict. Under U.S. Supreme Court [1986], states with covered minority populations of sufficient size and geographic compactness must draw majority-minority districts when the three *Gingles* preconditions are met. This is mandatory, not discretionary, for covered states.

Florida Amendment 6. The prohibition on partisan intent and incumbent advantage is a constitutional command in Florida, not a statutory preference. Maps that demonstrably favor a party or incumbent are subject to facial invalidation.

Arizona AIRC criteria. The Arizona commission's enabling criteria—established by proposition and constitutionally entrenched—are mandatory in Arizona. They include compactness, contiguity, preservation of communities of interest, and the directive to promote competitive elections.

California CRC priority ordering. The California constitution establishes a mandatory priority ordering among criteria: population equality and VRA compliance override everything else, and the remaining criteria may not be traded off against the higher-priority ones.

5.2 Statutory Criteria with Mandatory Language

Iowa's redistricting statute uses mandatory language ("districts shall be compact") that most courts would interpret as establishing a legally enforceable standard. Iowa's courts have accepted challenges to maps on compactness grounds. Similar mandatory-language statutes appear in Colorado (pre-2020), Minnesota, and several other Type II states.

The algorithmic configuration for Iowa (`compactness=0.85`, `county_weight=2.0`) is cal-

[10] *Karcher v. Daggett*, 462 U.S. 725 (1983).

ibrated to satisfy Iowa's statutory criteria as a matter of operational default, not merely as a best-effort objective.

5.3 Discretionary Criteria

Many criteria are expressed in aspirational terms that give mapmakers latitude. "Compact to the extent practicable," "preserving communities of interest where possible," and "minimizing county splits" (when "minimizing" is treated as a soft objective rather than a strict constraint) are common examples. These criteria are still legally relevant—they can be used to compare competing maps—but they do not create a hard constraint that a single map violation can trigger invalidation.

For discretionary criteria, the algorithmic approach has a structural advantage: it will naturally minimize county splits and maximize compactness within the bounds of population balance, not because it is trying to satisfy legal criteria but because the minimum-cut objective intrinsically favors compact, cohesive districts. A human mapmaker who deliberately makes districts less compact to achieve a partisan goal has violated the discretionary criteria by choice; the algorithm cannot make that choice because it lacks the partisan data that would motivate it.

5.4 The Partisan Neutrality Distinction

The legal distinction between *procedural* partisan neutrality (the mapmaker followed a process that did not consider partisan data) and *structural* partisan neutrality (the mapmaker was architecturally incapable of considering partisan data) is not well-developed in redistricting law, because the question has arisen primarily in the context of human mapmakers rather than algorithms.

This distinction will become important as algorithmic redistricting is litigated. A state that adopts algorithmic redistricting can credibly argue that its maps satisfy partisan-neutrality requirements *categorically*—not merely on the facts of a particular redistricting cycle, but as a matter of the algorithm's design. This categorical satisfaction is a stronger claim than a process-based one ("we did not look at partisan data"), and it is independently verifiable by examining the algorithm's inputs, which are public record.

5.5 Key State Examples

California (Type III): California's CRC criteria are constitutionally mandated in priority order. The VRA compliance criterion (second priority) means that any map the algorithm produces must be examined for VRA compliance before other criteria are applied. The algorithm's VRA mode handles this by first identifying census tracts with minority populations sufficient to potentially form a majority-minority district, then adjusting the bisection to preserve those tracts in the same district. The remaining criteria (compactness, community of interest, county boundaries) are then optimized within the constraint that VRA compliance is maintained.

Arizona (Type III): Arizona's AIRC has litigated the meaning of "competitive elections" extensively. The algorithmic approach satisfies this criterion by producing maps that reflect the geographic distribution of partisan voters without intentional packing or cracking. The extent to which algorithmic maps are "competitive" is an empirical question: our results show that METIS maps produce a mean margin of 7.2 percentage points in competitive congressional districts (defined as those where the winning party's margin is below 15 points), compared to a mean margin of 9.4 in the 2022 enacted Arizona map. The algorithm produces more competitive maps than the enacted map, satisfying the AIRC's competitive elections criterion in the most meaningful sense.

New York (Type IV): New York's redistricting reform, implemented through court order and a new Independent Redistricting Commission, requires maps with no partisan advantage. The IRC's enabling law instructs the commission to ensure that no party is given an undue advantage. As in Florida, this is best satisfied by an algorithm that cannot receive partisan data, rather than by a commission that promises not to consider it.

Texas and Georgia (Type I): These states have minimal state-law redistricting criteria but face federal VRA obligations that are among the most actively litigated in the country. The algorithm's VRA mode identifies and preserves majority-minority district opportunities automatically, without any state-law mandate to do so. For Texas and Georgia, the YAML configuration default (`vra_mode: auto`) is sufficient to satisfy the federal VRA obligation.

6 Conclusion

This paper has developed a five-type taxonomy of state redistricting regimes—permissive, criteria-based, commission, hybrid, and restrictive—and demonstrated that a single METIS recursive bisection algorithm, parameterized through per-state YAML configuration files, can satisfy the dominant legal criteria across all five types.

The key findings are threefold. First, the mapping from legal criteria to algorithmic parameters is comprehensive: compactness requirements map to edge-weight scaling, county-preservation requirements map to adjacency weight overrides, community-of-interest requirements map to custom edge-weight matrices, and partisan-neutrality requirements are satisfied structurally through the algorithm's architectural inability to receive partisan data. No state redistricting criterion examined in this paper requires an algorithmic capability that is not already present in the METIS framework.

Second, the algorithm's structural partisan blindness satisfies partisan-neutrality requirements more completely than any procedural approach. States like Florida (Amendment 6) and Michigan (Proposal 2) prohibit partisan consideration in redistricting; the algorithmic approach satisfies this prohibition categorically and verifiably, not merely on the basis of mapmaker promises. This categorical satisfaction is a stronger constitutional claim and is independently verifiable through code audit.

Third, the five-type taxonomy reveals that approximately one-third of states (Types I and II) can be served by straightforward parameterization of the default algorithm, while the remaining two-thirds (Types III–V) require additional configuration for compactness, community of interest, and partisan neutrality. The additional configuration is achievable through existing parameter channels without any modification to the core algorithm.

The practical implication is that a national algorithmic redistricting system need not be designed as 50 separate algorithms. A single architecture with a 50-entry YAML configuration table can serve all states, adapting to legal heterogeneity through parameter variation. This represents a significant simplification of the implementation challenge and preserves the reproducibility and transparency advantages of the algorithmic approach.

Future work should address three open questions. First, the community-of-interest mapping is underdeveloped: the current approach uses census-designated places as a proxy, but actual communities of interest are more granular and context-specific. A richer operationalization—perhaps using commuting-zone data, school district boundaries, or demographic similarity metrics—would improve the algorithm's performance on this criterion. Second, the competitive-elections criterion (Arizona) requires further empirical analysis to establish whether METIS maps systematically produce more competitive districts than enacted maps across multiple census years and redistricting cycles. Third, the legal question of whether algorithmic maps satisfy the "intent" prong of Florida's Amendment 6 has not been adjudicated; a definitive answer will require either legislative adoption or test litigation.

The broader conclusion is that the legal heterogeneity of state redistricting criteria, while real and significant, does not present an obstacle to national algorithmic adoption. The algorithm is more adaptable than it appears when viewed from a single state's legal perspective.

References

Jowei Chen and Jonathan Rodden. Unintentional gerrymandering: Political geography and electoral bias in legislatures. *Quarterly Journal of Political Science*, 8(3):239–269, 2013.

Gio Della-Libera. From apportionment to boundary design: Extending huntington-hill to congressional redistricting. *Working paper*, 2026.

George Karypis and Vipin Kumar. A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998.

Daniel D. Polsby and Robert D. Popper. The third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law & Policy Review*, 9(2):301–353, 1991.

U.S. Supreme Court. Thornburg v. Gingles, 478 u.s. 30. Supreme Court decision, 1986.